HOMEWORK ASSIGNMENT 8

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

QUESTION 1: Rent Comparisons

Are apartments priced differently in San Luis Obispo versus Santa Barbara? To answer this question, we'll take a look at a dataset containing 362 different apartments and their **city** (SLO or SB) and their **rent** (monthly rent in USD), as sourced from *Zillow*[1].

(a) What is one observational unit in the context of this dataset?

(b) Identify the explanatory and response variables, and classify each as either categorical or quantitative.

(c) Write the null and alternative hypotheses in symbols and math only, taking care to define all symbols you use.

Note: the remainder of this problem will require you to have copy-pasted the data into the Multiple Means Applet. Please follow the instructions outlined in Lecture Activity 16 ("Plate Up!") to accomplish this.

The data can be accessed using this link.

(d) Fill in the following table of summary statistics from the dataset.

San Luis Obispo	Santa Barbara
$n_{SLO} =$_____	$n_{SB} =$_____
$\bar{x}_{SLO} =$_____	$\bar{x}_{SB} =$_____
$s_{SLO} =$_____	$s_{SB} =$_____

Table 1: Summary Statistics

(e) Calculate the value of the statistic, using the order (SLO − SB).

(f) Use at least 10,000 repetitions to obtain an approximate simulation-based p-value. **Note:** there may be a slight pause between when you click "Shuffle Responses" and when the applet displays the simulated null distribution, because the dataset is somewhat large.

(g) Interpret the p-value you obtained in Part (f) above.

(h) Evaluate this p-value: How much evidence do the sample data provide against the null hypothesis? Choose one.

- ◯ Not much evidence against the null hypothesis
- ◯ Moderate evidence against the null hypothesis
- ◯ Strong evidence against the null hypothesis
- ◯ Very strong evidence against the null hypothesis

[1] Data is current as of October 20, 2026, but may contain apartments outside of the stated cities. The data has also not been thoroughly vetted.

(i) To what population can we generalize the results of this study? Make the best argument you can with the provided data, but also comment on any additional information you think would be pertinent to have.

(j) Given the results of your analysis, provide an overall conclusion in context. Be very clear what you can claim about the population means. IF you have evidence that one population mean is greater/smaller/different than the other, then you need to state which one is greater/smaller/different.

(k) Explain why we are or are not justified in using a theory-based approach to answer the research question. **Hint:** you may need to go back to the applet and get some additional information about the data!

(l) Critique this study. In particular, comment on how well you think the study is actually addressing the research question.

(m) Return to the applet and check the "95% CI(s) for difference in means" button, and report the resulting confidence interval. Also interpret this interval.

QUESTION 2: Height and Handedness

Do right-handed and non-right-handed US adults[2] have different heights? To answer this question, we'll use our data from the Welcome Survey. As a reminder: I sent out the welcome survey to my three STAT 1110 sections. After cleaning up the data (removing missing values, correcting mis-inputs, etc.) I was left with 82 responses. We'll consider the two variables to be "**handendess**" (with values "right-handed" or "not right handed") and "**height**" (as measured in inches). Here is the summary of the data:

Right-Handed	Not Right-Handed
$n_{\text{Right}} = 67$	$n_{\text{NotRight}} = 15$
$\bar{x}_{\text{Right}} = 65.984$ in	$\bar{x}_{\text{Right}} = 64.850$ in
$s_{\text{Right}} = 2.823$ in	$s_{\text{NotRight}} = 3.652$ in

Table 2: Summary Statistics

Additionally, we'll assume the variable "height" has a symmetric distribution regardless of a person's handedness, which will allow us to proceed with a theory-based approach to inference even though one group has fewer than 20 observations.

Whenever necessary, use the order (Right $-$ NotRight).

(a) Write down the null and alternative hypotheses in symbols and math only.

(b) Provide a verbal description of any symbols you used in Part (a) above.

[2] For this problem, we'll classify anyone college-age or older to be an "adult"

(c) Use the Theory-based Inference applet to carry out a two-sample *t*-test to answer the research question. To show your work include a screen shot of your output showing the (1) null distribution (curve), (2) standardized statistic and (3) p-value. Save the screen shot as a picture and upload it to this question.

(d) Should we reject or fail to reject the null hypothesis?

- ◯ Reject the null hypothesis in favor of the alternative hypothesis
- ◯ Fail to reject the null hypothesis

(e) Navigate back to the Theory-Based Inference applet, and construct a 95% confidence interval for the difference in population average heights. Interpret your interval.

(f) A student has made the following claim:

> Because the sample (STAT 1110 students in Dr. Ethan's sections) was not obtained through random sampling, we are not able to make cause and effect claims.

Do you agree with the student's reasoning? Explain.